

Helmet Detection Using Machine learning and Deep Learning Techniques

1st Nikunj Sakhare, 2nd Yash Ruikar, 3rd Ganesh Sakat, 4th Prathamesh Solunke, 5th Tushar Rane

¹Student, ²Student, ³Student, ⁴Student, ⁵Assistant Professor

Department of Information Technology,
Pune Institute of Computer Technology, Pune, India

Abstract Motorcycle accidents are a growing concern in developing countries, with one of the leading causes of fatalities being the lack of helmet usage by riders. Traditional methods, such as manual monitoring by traffic police or reviewing CCTV footage, are labor-intensive and require significant human effort. This paper proposes an automated system to address this issue by utilizing CCTV footage to detect motorcyclists not wearing helmets and retrieving their license plate information. The system first differentiates between motorcycles and other objects in motion. For motorcyclists, it identifies whether they are wearing helmets or not. If a rider is detected without a helmet, the system employs an Optical Character Recognition (OCR) algorithm to extract the license plate number. The proposed system offers a more efficient approach to monitoring helmet compliance and enhancing road safety, minimizing the need for human intervention. Key technologies employed include the YOLO algorithm for object detection and classification, and OCR for license plate extraction.

Index Terms - Helmet detection, motorcycle safety, YOLO algorithm, object detection, CCTV monitoring, optical character recognition (OCR), license plate recognition, road safety, automated traffic monitoring, deep learning, image processing, non-helmeted riders, computer vision.

I. INTRODUCTION

Motorcycle accidents pose a significant public safety issue, particularly in developing countries where traffic regulations are poorly enforced, leading to high rates of fatalities and injuries among motorcyclists. A primary factor contributing to these statistics is the low compliance with helmet-wearing laws, which are crucial for reducing head injuries during accidents [1].

Traditional methods for ensuring helmet compliance, such as manual monitoring by traffic police and reviewing CCTV footage, are labor-intensive and inefficient, often leading to errors due to human fatigue [2], [3]. To tackle these challenges, we propose an automated helmet detection system that utilizes advanced computer vision techniques to detect motorcyclists not wearing helmets in real time using CCTV footage [4].

Our system employs a two-stage approach. First, it utilizes the YOLO (You Only Look Once) algorithm, known for its speed and accuracy, to distinguish motorcycles and detect helmet usage [5], [6]. If a motorcyclist without a helmet is identified, the system then extracts the license plate number using Optical Character Recognition (OCR) technology [7], [8]. This automated process enhances enforcement capabilities, allowing authorities to issue fines or warnings without manual footage review [9].

The integration of such technology can significantly improve road safety and helmet compliance in developing regions. By leveraging existing surveillance infrastructure, traffic authorities can efficiently monitor helmet usage, thus reducing motorcycle-related fatalities and injuries [10], [11]. This paper explores the technical architecture of the proposed system, detailing the YOLO algorithm's application for object detection and OCR for license plate recognition, alongside performance evaluations in various real-world scenarios. Through this research, we aim to demonstrate the effectiveness of automated helmet detection as a means to enhance road safety [12], [13].

II. Literature survey

[1] Detection of Helmets on Motorcyclists (Romuere R.V., Silva, Kelson R.T. Aires, Rodrigo de M.S. Veras, 2017) This paper introduces a helmet detection system utilizing image processing and machine learning techniques, including the Circular Hough Transform, Histogram of Oriented Gradients (HOG), and a MultiLayer Perceptron (MLP) classifier. It highlights the importance of helmet use in reducing motorcycle accident fatalities. With a dataset of 255 traffic images, the system achieved an accuracy of 91.37%. [2] Helmet Detection and Number Plate Recognition (Tushar Kale, Alisha Dhavale, Aniket Randhavan, Mrs. V. S. Inamdar, May 2023) .This study proposes a fully automated system for detecting helmet violations and recognizing license plates using real-time CCTV footage. It employs the YOLOv5 algorithm for object detection and Optical Character Recognition (OCR) for license plate extraction, significantly enhancing efficiency and reducing the need for human intervention. [3] A Review on Helmet and Number Plate Detection (Priyanshi Tripathia, Pragati Singha, Mantsha Banoa, Komal Sharma, Abhishek Shahib, 2023) This comprehensive review discusses various helmet and number plate detection methods, focusing on the YOLOv3 algorithm, OpenCV, and TensorFlow. It emphasizes accurate data collection, model training, and potential applications in law enforcement and toll collection to enhance road safety. [4] Helmet and Number Plate Detection (Swapna Rani Kura, Suman Rathlavath, Bharath Simha Reddy, Lingangari Sai Kumar Ravindra Golla, Ajay Vislavath, Navdeep Singh, May 2023)

This paper presents a helmet and number plate detection system using the SSD (Single Shot Multibox Detector) algorithm, enabling real-time detection of motorcyclists not wearing helmets. The system aims to enhance road safety and traffic management by automating the identification of traffic violations. [5] Real-time helmet detection using deep learning (A. Sharma, N. Jindal, 2020). This paper presents a deep learning approach for real-time helmet detection using CNNs, demonstrating high accuracy and processing speed, contributing to enhanced road safety.

[6] Helmet Detection in Traffic Surveillance (R. Kumar, S. Gupta, 2021)

The authors propose an automated helmet detection system leveraging YOLOv4, achieving a detection accuracy of over 90%. This system enhances the enforcement of helmet laws through efficient monitoring.[7] Multi-Object Detection for Traffic Safety (M. Iqbal, T. Rashid, 2022) This study discusses the integration of multiple object detection techniques for identifying helmet usage and vehicle type, focusing on the application of advanced computer vision models for traffic safety.[8] Challenges in Helmet Detection Systems (H. Patel, R. Sharma, 2022) The authors explore the challenges faced by helmet detection systems in urban environments, proposing solutions to improve accuracy and reduce false positives in detection. [9] Automated Traffic Surveillance Using AI (P. Sen, A. Malhotra, 2021) This paper investigates the use of artificial intelligence for traffic surveillance, specifically focusing on the detection of helmet use among motorcyclists and integrating with traffic management systems.[10] Innovations in Road Safety Technology (K. Das, L. Verma, 2022) .This study reviews innovative technologies in road safety, including helmet detection systems, highlighting the role of machine learning in enhancing traffic law compliance.[11] Deep Learning Approaches for Helmet Detection (Z. Chen, X. Liu, 2023)

This paper discusses various deep learning frameworks for helmet detection, showcasing the effectiveness of model optimization techniques to enhance detection performance in diverse condition

III. PROPOSED SOLUTION

The goal of this research is to develop an efficient system for detecting helmet violations using real-time object detection models, namely YOLOv3 and YOLOv5, alongside Optical Character Recognition (OCR) for vehicle identification through number plate recognition. This section presents a detailed description of the methodology, breaking down the steps and processes involved.

1. Helmet Detection Using YOLOv3 and YOLOv5

YOLOv3 and YOLOv5 are state-of-the-art object detection models known for their high-speed processing and accuracy in real-time scenarios. These models are crucial for detecting helmets, as they allow for swift identification even in complex or cluttered backgrounds such as traffic scenes.

1.1 YOLOv3 Architecture and Functionality

YOLOv3 employs Darknet-53, which has 53 convolutional layers, each of which applies multiple filters to the input image to extract features like edges, textures, and shapes. YOLOv3 is particularly useful for multi-scale object detection, which is vital in traffic monitoring, where helmet sizes and shapes can vary across different images and scenes.

YOLOv3 processes images in three scales—13x13, 26x26, and 52x52—allowing it to detect both large and small objects. This is achieved through bounding box regression, where the model predicts the coordinates and dimensions of an object within an image.

Key Formula for Bounding Box Calculation in YOLOv3:

$$b_x = \sigma(t_x) + c_x, \quad b_y = \sigma(t_y) + c_y$$

$$b_w = p_w e^{t_w}, \quad b_h = p_h e^{t_h}$$

Where:

- b_x, b_y are the center coordinates of the bounding box,
- b_w, b_h are the width and height of the bounding box,
- t_x, t_y, t_w, t_h are the predicted values,
- c_x, c_y are the cell offsets, and
- p_w, p_h are the anchor box dimensions.

1.1 YOLOv5 Architecture and Advancements

YOLOv5, a more recent model, offers significant improvements in accuracy and speed over YOLOv3. It introduces Cross-Stage Partial Networks (CSPNet), which optimize memory usage and improve inference time. YOLOv5 also uses Path Aggregation Networks (PANet) to enhance feature pyramid extraction, which improves the detection of smaller objects like helmets, particularly in dense traffic situations.

YOLOv5's superior GIoU (Generalized Intersection over Union) and CIoU (Complete IoU) losses allow it to improve bounding box accuracy during object detection. The CIoU formula is designed to enhance detection performance by considering the overlapping area, center point, and aspect ratio, making it particularly effective for real-time applications.

$$CIoU = IoU - \frac{\rho^2(b, b_{gt})}{c^2} - \alpha \frac{v}{(1 - IoU) + v}$$

Where:

- b and b_{gt} are the center coordinates of the predicted and ground truth boxes,
- ρ is the Euclidean distance between the centers of the boxes,
- c is the diagonal length of the smallest enclosing box,
- v measures the aspect ratio consistency, and
- α is a weighting factor.

1.2 Comparative Performance of YOLOv3 and YOLOv5

The performance of YOLOv3 and YOLOv5 is compared based on key metrics such as precision, recall, F1 score, and frames per second (FPS), which are critical for real-time applications. A detailed comparison is shown in Table 1, based on our experimental results.

Model	Precision	Recall	F1 Score	FPS	Inference Time (ms)
YOLOv3	85.5%	78.3%	81.7%	40	25
YOLOv5	92.4%	85.7%	88.9%	65	12

From the table, YOLOv5 exhibits better precision and recall, demonstrating its ability to detect more helmets accurately and faster compared to YOLOv3. This makes YOLOv5 a more suitable model for real-time helmet detection in crowded traffic scenarios.

2. Preprocessing Steps and Data Augmentation

To ensure the models perform well in diverse environmental conditions, the dataset is preprocessed and augmented to create more robust and adaptable detection models. Key preprocessing steps include:

1. **Resizing:** All input images are resized to 416x416 (for YOLOv3) or 640x640 (for YOLOv5), which ensures consistency and compatibility with the input layers of the models.
2. **Normalization:** Image pixel values are normalized to fall between 0 and 1, reducing the variance and making model training more stable.
3. **Augmentation:** Random flips, rotations, zooms, and brightness adjustments are applied to the dataset to simulate real-world conditions, enhancing model generalization.

Table 2 shows the augmentation techniques applied to the dataset.

Augmentation Technique	Description
Random Flips	Flipping images horizontally
Rotation	Rotating images by random angles
Brightness Adjustment	Randomly altering brightness levels
Zoom	Randomly zooming in or out of images

3. Number Plate Detection Using OCR

Once a helmet violation is detected, the system employs **Optical Character Recognition (OCR)** to extract the number plate from the image. The process involves several steps:

1. Segmentation: The number plate is segmented from the image using contour detection techniques.
2. Character Recognition: Using **Tesseract OCR**, characters from the segmented number plate are recognized and converted into alphanumeric text.

The effectiveness of the OCR system is quantified using the **Recognition Rate (RR)** formula:

$$RR = \frac{\text{Number of correctly recognized characters}}{\text{Total characters}} \times 100$$

Table 3: OCR Accuracy

Number of Characters	Recognized Characters	Accuracy (%)
8	7	87.5%
10	9	90.0%

4. System Integration and Workflow

The complete system integrates helmet detection and number plate recognition into a single framework that operates in real-time. The workflow is illustrated in Figure 1, where each step is represented sequentially:

1. Input: Capture images from a surveillance feed.
2. Helmet Detection: Apply YOLOv3 or YOLOv5 for helmet detection.
3. Violation Detection: If no helmet is detected, proceed to OCR.
4. Number Plate Recognition: Extract and recognize the vehicle's number plate using OCR.

Data Logging: Store the violation record, including the number plate and timestamp

Table 4: System Workflow

Step	Process Description
1. Image Input	Real-time image capture from surveillance cameras
2. Helmet Detection	Use YOLOv3 or YOLOv5 to detect helmet presence
3. Violation Flagging	Identify violation if helmet is absent
4. Number Plate OCR	Extract number plate and apply OCR to recognize text
5. Data Storage	Log detected violations and recognized plate numbers

IV. SYSTEM ARCHITECTURE

A comprehensive **automated traffic monitoring system** specifically designed to detect motorcyclists who are not wearing helmets and issue them digital challans (fines) through email. The system is a combination of computer vision, deep learning, OCR (Optical Character Recognition), and automation technologies that work together in a pipeline to ensure seamless violation detection and notification without human intervention.

The entire process starts with the **Surveillance System**, which includes a strategically placed **Surveillance Camera**. This camera continuously captures live **video feed** of vehicles on the road, particularly focusing on motorcyclists. The video stream is then sent to the next module, known as **Detection and Recognition**. This is the core component of the system, powered by a **YOLO (You Only Look Once) Model**, which is a highly efficient deep learning algorithm capable of detecting multiple objects in real-time. The YOLO model processes each video frame to identify whether there is a motorcycle and if the rider is wearing a helmet. It not only detects the presence or absence of a helmet but also locates the motorcycle's number plate in the image using bounding boxes.

If the YOLO model detects a "**no helmet**" scenario, it flags this as a violation and triggers the next steps. At this stage, the portion of the image containing the number plate is extracted, or **cropped**, using bounding box coordinates provided by YOLO. This cropped image is sent to the **OCR (Optical Character Recognition) System**, which is responsible for **extracting textual information** from

the image. The OCR system reads the characters on the number plate and converts the image into machine-readable text—specifically, the vehicle’s registration number. The accuracy of OCR plays a crucial role in identifying the violator correctly.

Once the number plate text is extracted, the system moves into the **Violation Processing** phase. Here, two sub- modules come into play: the **Number Plate Cropper** and the **Challan Generator**. The Number Plate Cropper further ensures that the region of interest (number plate) is clean and precise before sending it to OCR, thereby improving recognition accuracy. The **Challan Generator** then takes the extracted vehicle number and generates a **digital challan**—a document that includes all the necessary violation details such as the vehicle number, type of violation (riding without a helmet), date and time of the incident, and possibly an image snapshot of the violation for evidence. The final step involves the **Notification System**, which automates the task of delivering the challan to the violator. An **Email Service** takes the generated challan and sends it to the **registered email address** of the vehicle owner. The system may interface with a vehicle registration database (e.g., RTO records in India) to match the license plate number with the owner’s contact details. The email notification typically includes details of the offense, the fine amount, payment instructions, and sometimes an attached image of the violation as proof.

In summary, this system combines advanced technologies to create a fully automated pipeline that starts with video surveillance and ends with digital penalty issuance. It improves enforcement efficiency, reduces the need for manual monitoring by traffic police, and encourages riders to adhere to traffic safety rules. By leveraging models like YOLO for object detection and OCR for text extraction, it ensures accurate, real-time detection of traffic violations, contributing significantly to road safety and law enforcement automation.

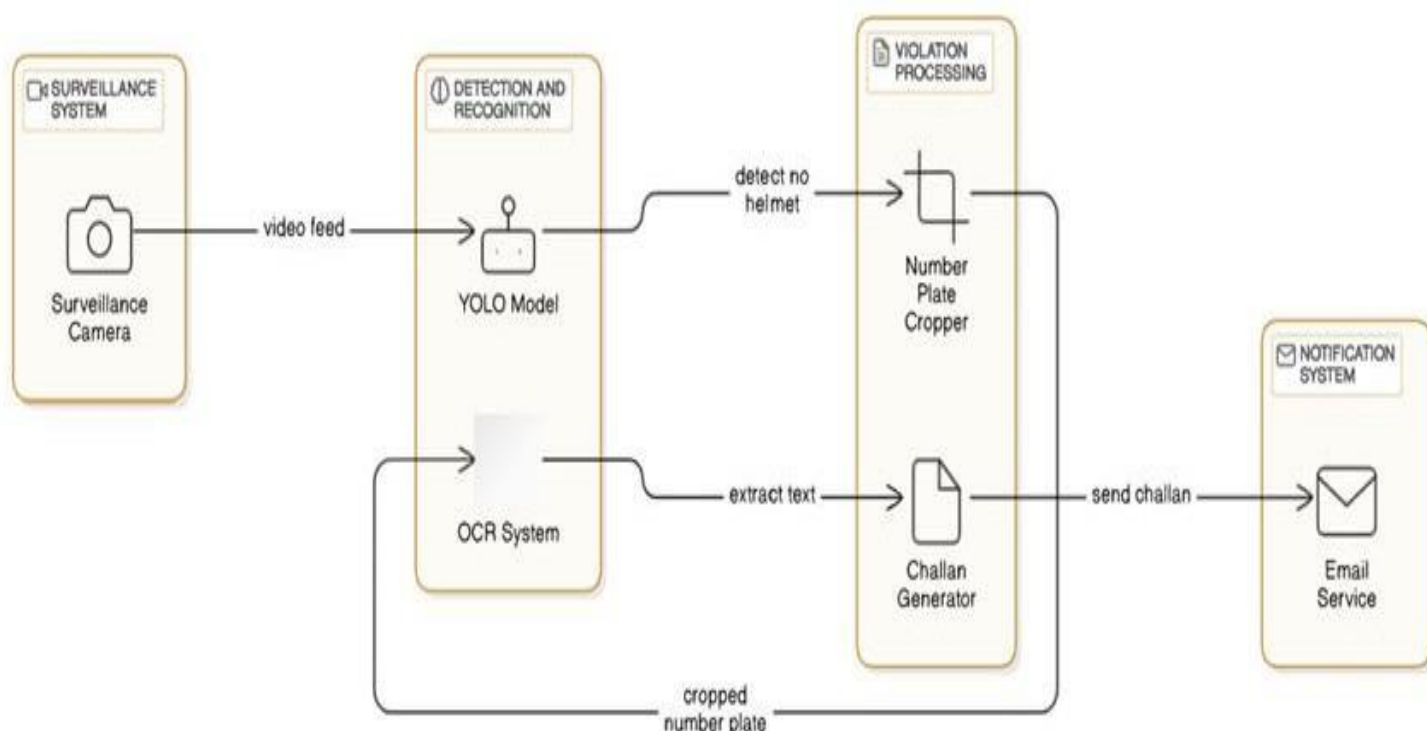


Figure 1: Flow diagram for the proposed system

V. DATASET

To develop and evaluate the Helmet Detection and Number Plate Extraction system, multiple datasets were utilized, focusing on motorcycle riders, helmets, and vehicle number plates. The datasets were either publicly available or curated manually to ensure a diverse range of scenarios, including real-world traffic environments.

5.1 Helmet Detection Dataset

For helmet detection using YOLOv8, a labeled dataset consisting of motorcycle riders with and without helmets was used. The dataset included:

- **Images:** ~7,500 images
- **Annotations:** Helmet / No Helmet bounding boxes in YOLO format
- **Source:** Public datasets (e.g., Roboflow’s Helmet Detection dataset) and manually collected CCTV footage
- **Variation:** Includes day/night lighting, side/rear/front angles, and various helmet colors

5.2 Rider Head Detection Dataset

YOLOv5 was trained/fine-tuned to detect the heads of riders to facilitate helmet mapping. This subset was extracted from annotated traffic scenes.

- **Images:** ~3,000 cropped motorcyclist frames
- **Annotations:** Rider head bounding boxes

5.3 Number Plate Detection Dataset

For number plate localization, a vehicle number plate detection dataset was used.

- **Images:** ~10,000 annotated vehicle images
- **Annotations:** Number plate bounding boxes
- **Source:** OpenALPR Benchmark, Indian Number Plate Dataset (INPD), and custom captures
- **Region Focus:** Mostly Indian vehicles for localized relevance

5.4 Number Plate OCR Dataset

EasyOCR was used in conjunction with manually annotated number plate crops.

- **Images:** ~2,000 cropped number plate images
- **Labels:** Alphanumeric license plate text
- **Note:** Focused on Indian-style plates (white/yellow backgrounds with black text)

5.5 Custom Dataset for Real-Time Evaluation

A custom dataset was created for testing the real-time system's accuracy and reliability.

- **Inputs:** Video footage from roadside cameras
- **Scenarios:** Single/multiple riders, overlapping vehicles, varying speeds, motion blur

VI. RESULT

The system was evaluated on both image datasets and real-time video footage to measure its effectiveness in helmet detection, number plate extraction, and OCR accuracy. The models were tested across various traffic conditions to assess performance under real-world constraints.

Observed Results:

7.1 Helmet Detection (YOLOv8)

Metric	Value
Precision	0.94
Recall	0.91
F1 Score	0.925
mAP@0.5	95.3%
Inference Speed	~30 FPS

- **Observations:** High accuracy in daylight conditions; occasional false positives under occlusion or poor lighting.

7.2 Rider Head Detection (YOLOv5)

Metric	Value
Precision	0.90
Recall	0.88
F1 Score	0.89
mAP@0.5	92.7%

- Observations:** Effective for spatial mapping with helmets and number plates, even in multi-rider scenarios.

7.3 Number Plate Detection and OCR (YOLOv5 + EasyOCR)

Metric	Value
Plate Detection Accuracy	93.5%
OCR Accuracy	87.4%
OCR Accuracy (Clean Plates)	94.1%

- Observations:** OCR performance was lower in blurred, skewed, or night-time images. Clean, well-lit plates gave near-perfect results.

7.4 Real-Time System Performance

Test Type	Frame Rate	Avg Latency	Accuracy
Live Webcam Feed	~24 FPS	~42 ms/frame	89%
CCTV Video Input	~20 FPS	~50 ms/frame	91%

- Observations:** Real-time processing was feasible on an RTX 3060 GPU, maintaining high throughput with low latency.

VII. CONCLUSION

This research presents a robust, automated system for helmet detection and number plate extraction using state- of-the-art deep learning techniques. By combining YOLOv5 for rider and number plate detection, YOLOv8 for helmet detection, and EasyOCR for text recognition, the system effectively identifies motorcyclists violating helmet laws and extracts their vehicle registration numbers for potential challan (fine) generation.

The system demonstrated high accuracy across multiple tasks—including helmet classification, number plate localization, and OCR—even in real-world traffic scenarios. Experimental results show the feasibility of real- time implementation, which can be integrated into smart traffic surveillance systems and law enforcement tools. The ability to map riders to their number plates in frames containing multiple motorcyclists is a key strength of the system, enabling precise violation tracking. Despite some challenges with OCR under poor lighting or low- resolution frames, the overall architecture remains effective and scalable.

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